Beyond Random Warps: Family-Level Estimation for Transferable Deformation Attacks

Anonymous Author(s)

Abstract

Deformation-based transferable attacks are usually strengthened by sampling more warped views, yet DeCoWA-style methods still estimate the update from one stochastic warp family. Under the released DeCoWA 3×3 TPS mesh, the learnable interior warp variable is effectively two-dimensional, which makes structured coverage of deformation families feasible. We therefore propose **MFC** (*Multi-Family Consensus*), a family-level estimator that averages gradients within random, equiangular, and antithetic warp families and fuses only the resulting family directions. Our primary operating point, **MFC-Std**, is a DeCoWA-compatible redesign; **MFC-Full** asks whether the same factorization still helps after replacing the outer search with OPS. On matched 1000-image ImageNet transfer benchmarks with ResNet-50 and ViT-B/16 surrogates, MFC-Std improves over DeCoWA by 13.1 and 5.4 black-box-ASR points, while MFC-Full improves over matched OPS by 2.1 and 2.8 points. Complementary fixed-budget controls show that a three-family 20×3 setting still exceeds a 60-random-warp baseline by 2.2 black-box-ASR points, and the strongest generic 2-D sampler remains 0.7 points behind. The largest MFC-Std gains occur under stronger source-target mismatch, especially from a CNN surrogate to transformer-like targets. These results establish family-level estimation as a strong deformation-attack operating point and show that the gain is not reducible to extra random views or a single better 2-D sequence.

CCS Concepts

• **Security and privacy** → **Software and application security**;
• **Computing methodologies** → *Computer vision.*

Keywords

transferable adversarial attacks, black-box robustness, deformation-constrained warping, gradient estimation, ImageNet transfer

ACM Reference Format:
Anonymous Author(s). 2026. Beyond Random Warps: Family-Level Estimation for Transferable Deformation Attacks. In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26), November 10–14, 2026, Rio de Janeiro, Brazil.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Transferable adversarial attacks remain one of the most realistic black-box threat models for modern visual systems because an attacker can optimize on a released surrogate and still affect unseen victims [2, 4, 9, 11, 17, 29]. We study this phenomenon at the shared image-side backbone, where the central question is whether an update direction survives model mismatch at the classifier layer.

MM '26, Rio de Janeiro, Brazil
2026. ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Recent progress has enlarged the surrogate neighborhood through transformation stacks, multi-stage attack pipelines, feature-aware guidance, surrogate regularization, and broader operator search [1, 3, 5–8, 12, 14, 20–25, 28, 30, 32, 34]. These directions are often effective and complementary, but they also concentrate methodological effort on increasingly elaborate outer pipelines. A simpler question is easier to overlook: within a fixed attack backbone, is the estimator used to aggregate transformed views actually aligned with the geometry of the view space? This distinction matters because stronger transfer can reflect either a better direction estimate or simply a larger surrogate-side search budget.

Deformation-based attacks provide a clean test case for this underexplored axis. Methods such as SIA, BSR, L2T, DeCoWA, and OPS already suggest that how views are organized matters at least as much as how many views are sampled [5, 10, 20, 23, 35], yet the dominant response has still been to enrich view generation or strengthen the outer pipeline. The released DeCoWA implementation makes the missed opportunity unusually concrete: under the default 3×3 TPS mesh, there is only one interior control point, so the operative warp variable is effectively two-dimensional. In that regime, the estimator itself becomes the main design object, yet prior deformation-based attacks do not turn this low-dimensional structure into an explicit estimator-design problem.

This observation motivates MFC as a principled minimalist redesign inside an existing backbone. We do not introduce a new loss, alter the surrogate, or add another transformation type. Instead, we replace unstructured single-family random sampling with a structured estimator that first stabilizes gradients within complementary deformation families and only then fuses the resulting family directions. In this setting, the simplest design change that no prior method made is often the most revealing one: it tests whether transfer is limited by estimator organization before invoking a larger outer pipeline.

Concretely, MFC constructs random, equiangular, and antithetic deformation families, averages gradients within each family, and then mixes the resulting family directions. The DeCoWA-backed operating point, MFC-Std, isolates this estimator redesign inside the released deformation backbone, while the stronger MFC-Full row asks whether the same factorization remains useful after the outer search is upgraded to the released OPS backbone. Complementary fixed-budget controls already preview the answer: under matched 60-view computation, the three-family 20×3 estimator stays +2.2 above a 60-random-warp control, and the strongest generic 2-D sampler still trails by 0.7. The gain therefore points to a more informative organization of the same view budget rather than to more computation.

Across matched 1000-image TransferAttack benchmarks, the same conclusion holds at scale. MFC-Std improves over DeCoWA from 75.7% to 88.8% black-box ASR on the ResNet-50-source block and from 79.9% to 85.3% on the ViT-B/16-source block. MFC-Full further improves over matched OPS from 95.0% to 97.1% and from 93.9%

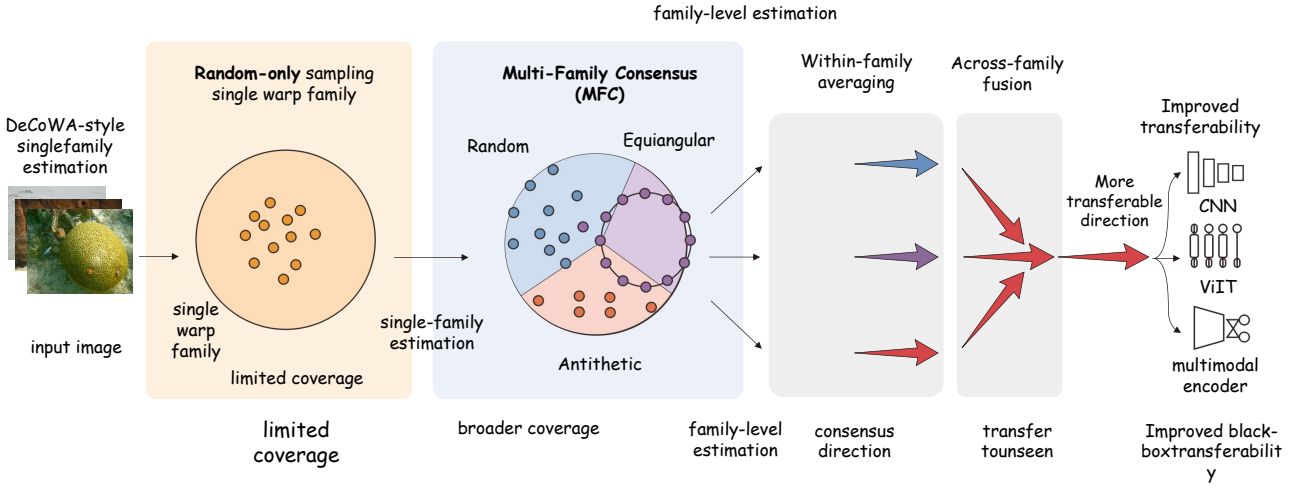


Figure 1: Motivation of MFC. DeCoWA-style single-family estimation samples one stochastic warp family, whereas MFC broadens coverage across complementary deformation families, averages gradients within each family, and fuses only the resulting family directions into a more transferable update.

to 96.7%, respectively. The largest MFC-Std gains appear when a CNN source attacks transformer-like victims.

Contributions.

- We identify an under-explored estimator-design axis in deformation-based transfer: under the released DeCoWA 3×3 TPS mesh, the operative warp variable is effectively two-dimensional, yet prior deformation attacks do not turn this low-dimensional structure into an explicit family-level estimation problem.
- We propose **MFC**, a principled estimator redesign that keeps the loss, surrogate, and deformation type fixed, replacing single-family random sampling with within-family stabilization and across-family fusion; we instantiate it as the controlled **MFC-Std** design plus the stronger OPS-backed **MFC-Full** generalization check.
- On matched ImageNet benchmarks and matched 60-view controls, MFC-Std outperforms DeCoWA, MFC-Full improves over OPS, and the three-family 20×3 estimator remains +2.2 above a 60-random-warp control while also outperforming the strongest generic 2-D sampler, with the largest MFC-Std gains under CNN-to-transformer transfer.

2 Related Work

2.1 Transfer Attacks by View Expansion

Iterative transfer attacks have progressively enlarged the surrogate neighborhood through momentum, input diversity, sample mixing, feature-aware guidance, ensemble variants, operator search, surrogate smoothing, and meta-surrogate training [1–3, 5–9, 12–15, 18, 20–24, 26–32, 34, 35]. Their implementations differ, but the shared idea is to query the surrogate through more informative views so that the resulting direction generalizes better to unseen targets.

Our work still belongs to this broad family, but changes the unit of expansion. We do not add an unrelated transformation zoo on top of an existing attack, nor do we retrain or smooth the surrogate itself. Instead, we take one strong view type, deformation-constrained warping, and ask how to organize its internal search geometry more systematically. This positions the paper as an estimator redesign inside one backbone rather than as another cross-attack branch stack.

This distinction is also what separates MFC from surrogate-space methods such as LRS and MTA [15, 27]. Those methods alter or regularize the surrogate so that its gradients become easier to transfer, whereas MFC keeps the surrogate fixed and changes only how deformation views are organized and fused. We therefore regard the two directions as conceptually orthogonal.

2.2 Deformation-Based Transfer

Geometric or structure-preserving view generation has become a distinct axis of transfer improvement. Local block-wise transformations, block shuffling, path augmentation, and deformation-constrained warping all attempt to expose gradients that survive architecture shift without destroying semantic content [10, 20, 23, 32]. DeCoWA is the closest backbone to our method because it already shows that deformation-constrained warping is effective across CNN and ViT families [10]. Our question is narrower: once deformation warping is fixed as the backbone, is one stochastic warp family still the right estimator of the deformation gradient? MFC answers this by replacing the single-family estimator with a structured family ensemble.

2.3 Evaluation Fairness

Transfer-attack comparisons are easy to distort through silent changes in source model, target family, step count, image subset, or evaluation protocol [33]. All reported numbers therefore come from released TransferAttack assets and matched local reruns on the same 1000-image split [16, 19], which keeps the estimator comparisons in Sec. 4 directly comparable.

3 Method

3.1 Preliminaries

We study the standard untargeted black-box transfer setting. Let f denote the surrogate classifier, x the clean image, y its ground-truth label, and ϵ the ℓ_∞ perturbation budget. The attacker seeks an adversarial example $x^{\text{adv}} = x + \delta$ such that

$$\|\delta\|_\infty \le \epsilon$$

and transfers to unseen target models. As in iterative transfer attacks, the perturbation is updated over T steps, and the practical design problem is to estimate a surrogate-side gradient that is less specific to the surrogate and more useful for transfer.

In the released DeCoWA pipeline, the key estimation step is performed in the warp space. Let $W(x_t; u)$ denote the thin-plate-spline warp used by DeCoWA [10], parameterized by a control variable u. In the public implementation, the interior noise map has shape $(\texttt{mesh_height} - 2, \texttt{mesh_width} - 2, 2)$ [19]; under the default 3×3 mesh this becomes $(1, 1, 2)$, so u is effectively a two-dimensional displacement vector. This low-dimensional geometry is the starting point of our redesign, and Figure 2 summarizes the resulting pipeline.

3.2 Single-Family versus Family-Mixture Estimation

The local redesign can be understood as a change in the estimator rather than a change in the outer attack objective. Let

$$g(u; x_t, y) = \nabla_x \mathcal{L}(f(W(x_t; u)), y)$$

denote the image gradient induced by warp parameter u. A single-family deformation attack estimates the transferable deformation direction through one family-specific sampling distribution $q(u)$:

$$\bar{g}_{\text{single}} = \mathbb{E}_{u \sim q(u)}\,[g(u; x_t, y)].$$

In the released DeCoWA-style setting, this family is stochastic and random. MFC replaces it with a mixture over family-specific warp distributions $q_m(u)$, where $m \in \mathcal{F}$:

$$\bar{g}_{\text{mix}} = \sum_{m \in \mathcal{F}} \pi_m \mathbb{E}_{u \sim q_m(u)}\,[g(u; x_t, y)],$$

where π_m is the family weight. In our implementation, the within-family expectation is approximated by $S = 20$ warped views. The weights depend on the operating point, but the defining change is invariant: first form one estimate per family, then fuse those family-level estimates.

3.3 A Bias-Variance View of Family Aggregation

Let $g_t^\star$ denote an unobserved transferable direction at attack step t. We write the family estimator as

$$g_{m,t} = g_t^\star + b_{m,t} + \xi_{m,t},$$

where $b_{m,t}$ is a family-specific directional bias and $\xi_{m,t}$ is a zero-mean stochastic term. For a single family, the expected squared error around the transferable direction becomes

$$\mathbb{E}\left[\|g_{m,t} - g_t^\star\|_2^2\right] = \|b_{m,t}\|_2^2 + \operatorname{tr}(\Sigma_{m,t}),$$

with $\Sigma_{m,t}$ the covariance of $\xi_{m,t}$.

For a weighted family mixture $g_t^{\text{mix}} = \sum_{m \in \mathcal{F}} w_m^t g_{m,t}$, the corresponding error is

$$\begin{aligned}\mathbb{E}\left[\|g_t^{\text{mix}} - g_t^\star\|_2^2\right] = &\left\|\sum_{m \in \mathcal{F}} w_m^t b_{m,t}\right\|_2^2 \\ &+ \sum_{m \in \mathcal{F}} (w_m^t)^2 \operatorname{tr}(\Sigma_{m,t}) \\ &+ 2\sum_{i<j} w_i^t w_j^t \operatorname{Cov}(\xi_{i,t}, \xi_{j,t}).\end{aligned}$$

This identity captures the design intuition: if family-specific biases partially cancel and the noise terms are not perfectly aligned, a family mixture can be closer to the transferable direction than any single family.

3.4 Three Deformation Families

We define a deformation-family set

$$\mathcal{F} = \{\texttt{random}, \texttt{equiangular}, \texttt{antithetic}\}.$$

For each family $m \in \mathcal{F}$, MFC constructs S initial warp parameters $u_{m,s}^{(0)}$, $s = 1, \ldots, S$, with $S = 20$ in all current runs. The families differ only in how these initializations are chosen.

Random family. Each $u_{m,s}^{(0)}$ is sampled independently from the released DeCoWA noise box. This family preserves the exploratory behavior of the original deformation attack and acts as the stochastic component of the estimator.

Equiangular family. Because the interior control space is effectively 2-D, directions can be placed uniformly on the circle. The s-th warp uses angle $2\pi s/S$ and the same nominal radius as the default noise scale. This family trades stochasticity for deterministic directional coverage.

Antithetic family. Warps are created in $\pm u$ pairs so that opposite directions are explored together. This family is designed to reduce one-sided sampling variance and to make the family estimator less sensitive to asymmetric local geometry.

Each initialization is refined by one DeCoWA-style inner update in the warp space:

$$u_{m,s} = u_{m,s}^{(0)} - \rho \nabla_u \mathcal{L}(f(W(x_t; u_{m,s}^{(0)})), y),$$

where ρ is the warp-space step size. This keeps the method compatible with the released DeCoWA pipeline while changing only the family organization.

3.5 Why These Three Families

The family identities are intentionally simple. The contribution of MFC is the estimator factorization instantiated through random, equiangular, and antithetic sampling, which play complementary roles in the released 2-D setting: random preserves exploration, equiangular supplies directional coverage, and antithetic pairing reduces one-sided variance. The last point can be seen from a local first-order expansion around a small direction u:

$$g(u; x_t, y) \approx g(0; x_t, y) + J_t u,$$

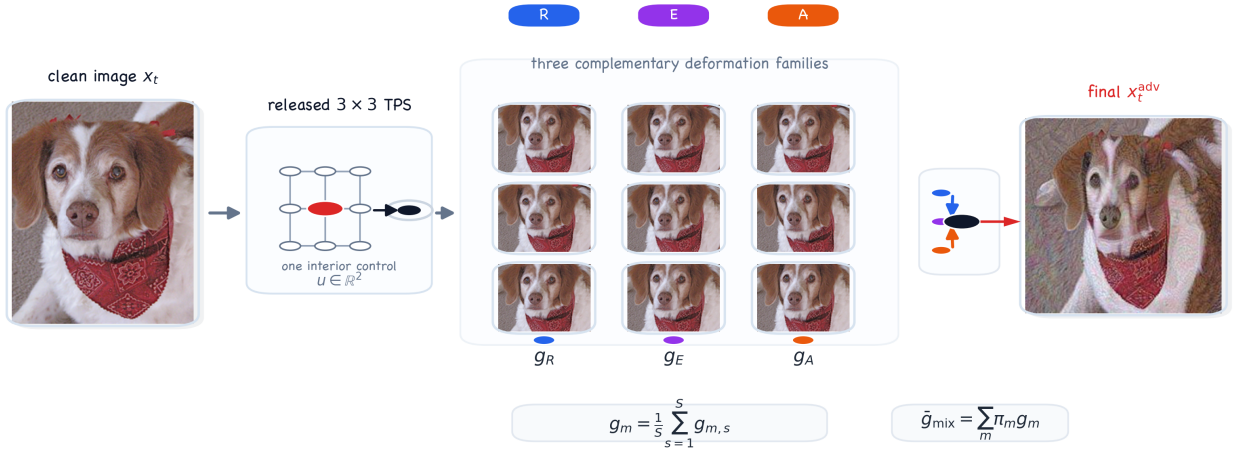


Figure 2: Overview of MFC-Std. In the released DeCoWA 3×3 TPS setting, MFC-Std builds three complementary deformation families, averages views within each family, and fuses only the resulting family directions in the final DeCoWA-compatible outer update.

where J_t is a local Jacobian term. The corresponding antithetic average satisfies

$$\frac{1}{2}\big(g(u; x_t, y) + g(-u; x_t, y)\big) \approx g(0; x_t, y),$$

which cancels the odd-order linear component and preserves the shared component around the center. This local view highlights why antithetic pairing can emphasize the common component around the center.

3.6 Within-Family Estimation

Once the refined warps are available, each family produces one family gradient by averaging image gradients over its warped views:

$$g_{m,t} = \sum_{s=1}^{S} \lambda_s \nabla_x \mathcal{L}(f(W(x_t; u_{m,s})), y).$$

In the current implementation, the within-family weights are uniform, so $\lambda_s = 1/S$. This choice keeps the method definition simple and makes the family comparison easy to interpret. If only the random family is active, the resulting estimator reduces to a DeCoWA-style deformation gradient inside the same TransferAttack API.

3.7 Across-Family Aggregation

The family gradients are then fused through a family-level mixture

$$g_t^{\text{fam}} = \sum_{i \in \mathcal{F}} w_i \cdot \frac{g_{i,t}}{\text{mean}(|g_{i,t}|)}, \qquad \sum_{i \in \mathcal{F}} w_i = 1, \; w_i \geq 0.$$

The crucial choice is that aggregation happens over *family-level* estimates rather than over individual warped views. A view-level rule would combine two different sources of variation: stochastic variation among samples from the same family and directional variation between genuinely different families. By averaging inside each family first, MFC separates variance reduction from cross-family selection and makes the ablation logic easier to interpret. In all paper-facing results we therefore use explicit fixed family weights rather than an adaptive consensus rule. The reduced family-only diagnostics compare equal, front-loaded, and balanced mixtures directly; the main MFC-Std row keeps the front-loaded mix used in the completed full runner, while the stronger MFC-Full row uses equal weights across OPS-side families.

3.8 Outer Update

The final **MFC** update uses the same outer MI-FGSM-style loop as the released runner:

$$\delta_{t+1} = \Pi_{\|\delta\|_\infty \leq \epsilon}\left(\delta_t + \alpha \cdot \text{sign}(m_{t+1})\right),$$

where m_{t+1} is the momentum update built from g_t^{fam}. The important difference from DeCoWA is therefore not a new loss, but a different estimator of the deformation gradient. MFC-Std keeps the DeCoWA-compatible outer runner, while MFC-Full keeps the same family-level factorization but delegates the outer neighborhood search to the released OPS backbone.

This compatibility matters for how we interpret the two paper-facing rows. MFC-Std is the controlled mechanism result because it changes only the deformation estimator inside the released DeCoWA backbone, whereas MFC-Full is a consistency check under a stronger outer search rather than a separate novelty claim.

4 Experiments

4.1 Experimental Setup

All numbers in this paper come from released TransferAttack models and matched local reruns under one protocol [19]. We use the standard ImageNet transfer setting [16], untargeted attacks, and perturbation budget $\epsilon = 16/255$. The main benchmark uses ResNet-50 as the surrogate and evaluates transfer to seven held-out targets: VGG16, MobileNet-v2, Inception-v3, ViT-B/16, PiT-B, Visformer-S, and Swin-T. We also report the corresponding white-box ASR on

the surrogate. Unless otherwise stated, the tables in this section report 1000-image reruns.

We use three summary metrics throughout this section. *Black-box Avg.* is the average ASR over the seven held-out targets. *CNN Avg.* and *Transformer Avg.* average the corresponding victim groups in each source block. For ResNet-50, this means {VGG16, MobileNet-v2, Inception-v3} versus {ViT-B/16, PiT-B, Visformer-S, Swin-T}; for ViT-B/16, it means {RN50, VGG16, MobileNet-v2, Inception-v3} versus {PiT-B, Visformer-S, Swin-T}. This split matters because a method that improves only one side of the target pool is less compelling than one that improves both, especially when the paper's main story is cross-architecture transfer.

The reported MFC-Std configuration keeps the DeCoWA-compatible deformation backbone, activates the three deformation families `random`, `equiangular`, and `antithetic`, and uses 20 warped views per family. Relative to DeCoWA's 20 random warps, MFC-Std therefore evaluates 60 deformation views per attack step; because each view incurs one warp-space update and one image-space gradient, this corresponds to 120 versus 40 equivalent gradient evaluations per step, i.e., an exact 3× increase in deformation-side compute. Later controls complement the main tables with exact 60-view comparisons, generic 2-D sampling alternatives, and a matched runtime profile.

The paper-facing MFC-Std row keeps the front-loaded family mix used in the completed full runner. Table 3 later compares equal, front-loaded, and balanced weightings directly inside a reduced family-only diagnostic. The stronger MFC-Full configuration applies the same family-level factorization after the outer search is upgraded to OPS. For that row, each OPS-side family mode keeps the published OPS(10, 30, 30) neighborhood settings inside its own family pool, and we then fuse three OPS-side family modes, `spatial`, `scale`, and `mixed`, with equal family weights. In the runtime profiler this corresponds to 2703 versus 901 equivalent gradient evaluations per step.

4.2 Main Benchmark: ResNet-50 Surrogate

Table 1 is the main result because it shows both levels of the paper claim under one matched benchmark: a controlled DeCoWA-backed redesign and a stronger OPS-backed generalization.

Table 1 supports two claims. First, within the DeCoWA backbone, MFC-Std raises black-box ASR from 75.7% to 88.8%, showing that the estimator redesign is already strong before any OPS machinery is introduced. Second, after replacing the outer search with OPS, MFC-Full still improves over matched OPS from 95.0% to 97.1%.

4.3 Where the Gain Appears

The black-box average is the cleanest single summary, but the target-wise columns clarify why the new story is stronger than a single-row win. Relative to DeCoWA, MFC-Std improves the CNN targets by +5.7 points on average (85.7% → 91.4%) but the transformer-like targets by +18.7 points (68.2% → 86.9%). The largest single-target gain is +28.7 on ViT-B/16. This concentration is exactly what one would expect from a better cross-architecture estimator rather than from a surrogate-only strengthening trick.

Figure 3 makes the same point at the per-target level. With a ResNet-50 source, the largest gains fall on transformer-like victims; after swapping the source to ViT-B/16, the gains remain positive but flatten, which is more consistent with a gap-driven explanation than with a source-only strengthening trick.

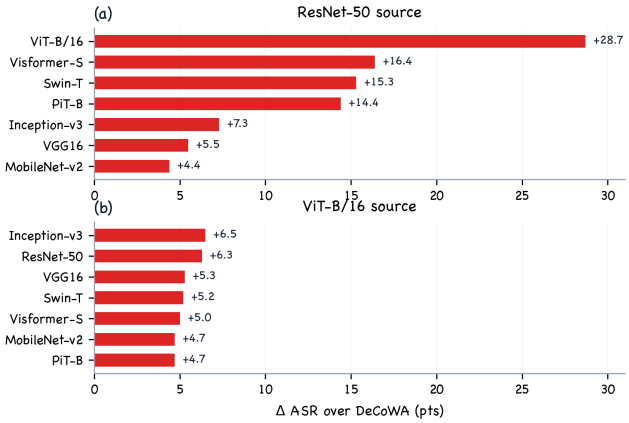


Figure 3: Per-target gains of MFC-Std over DeCoWA on the two completed source blocks. The largest deltas cluster under stronger source–target mismatch, especially when a CNN source attacks transformer-like victims.

Artifact diagnostics support the same reading: MFC-Std wins more paired examples than it loses on every held-out target, and on a 256-image architecture-gap analysis the gain over DeCoWA is 5.7 points on held-out CNN targets but 18.7 points on transformer targets, with source-target prediction agreement anticorrelation $r = -0.52$. The same table also separates transfer improvement from plain white-box saturation: compared with OPS, MFC-Full improves the white-box column by 1.0 point and the black-box average by 2.1 points.

4.4 Cross-Source Benchmark: ViT-B/16 Surrogate

The ViT-B/16-source block tests whether the same estimator story survives a source-family change. This is a harder generalization check than winning with one source because the deformation estimator is no longer paired with a CNN surrogate.

Table 2 shows that both claims remain intact after this swap. MFC-Std reaches 85.3%, improving over DeCoWA by 5.4 points, over BSR by 6.0 points, over SIA by 6.1 points, and over L2T by 2.8 points. MFC-Full reaches 96.7%, improving over OPS by 2.8 points, with gains on every held-out target and especially clear margins on PiT-B (+3.0), Visformer-S (+3.9), and Swin-T (+3.6). Together with the ResNet-50 block, this source swap shows that the estimator remains effective across surrogate families while preserving the strongest gains in the CNN-to-transformer regime highlighted in Table 1.

This source swap also broadens the empirical picture beyond one favorable surrogate. The gain gets smaller than in the ResNet-50 block, but it stays positive across the same matched protocol and therefore supports a source-robust estimator claim.

Table 1: Matched 1000-image transfer benchmark with a ResNet-50 surrogate. Family-level estimation yields a strong controlled operating point: MFC-Std gains +13.1 over DeCoWA, and MFC-Full adds +2.1 over matched OPS. All entries are ASR (%) ↑; *Black-box Avg.* averages the seven held-out targets.

Category	Attack	RN50 ↑	CNN targets			Transformer targets				Averages		
			VGG16	MN-v2	Inc-v3	ViT-B/16	PiT-B	Vis-S	Swin-T	CNN Avg. ↑	Transformer Avg. ↑	Black-box Avg. ↑
Gradient-based	VMI-FGSM	86.0	55.9	55.5	45.6	25.2	35.1	38.9	42.3	52.3	35.4	42.6
	GRA	90.5	75.9	76.0	73.4	44.6	56.8	63.0	65.0	75.1	57.4	65.0
Input-transformation-based	DIM	82.6	54.7	53.6	43.1	20.8	28.9	34.8	36.4	50.5	30.2	38.9
	SIM	88.9	54.9	53.1	40.6	17.8	26.9	33.8	36.0	49.5	28.6	37.6
	Admix	93.9	64.2	63.8	51.5	23.7	36.1	43.9	44.8	59.8	37.1	46.9
	SIA	95.3	83.1	84.1	69.2	38.2	57.7	67.9	67.4	78.8	57.8	66.8
	BSR	96.2	90.2	89.1	79.0	43.2	63.8	72.9	72.1	86.1	63.0	72.9
	DeCoWA	94.7	87.4	86.9	82.8	53.6	67.1	76.0	75.9	85.7	68.2	75.7
	L2T	95.6	87.0	87.6	80.9	53.4	68.2	74.5	74.6	85.2	67.7	75.2
	OPS	99.0	97.6	97.9	97.8	88.5	92.6	95.9	94.6	97.8	92.9	95.0
Ours	**MFC-Std**	97.7	92.9	91.3	90.1	82.3	81.5	92.4	91.2	91.4	86.9	88.8
Ours	**MFC-Full**	**100.0**	**98.9**	**99.3**	**98.1**	**92.3**	**95.5**	**97.4**	**98.2**	**98.8**	**95.9**	**97.1**

Table 2: Matched 1000-image cross-source transfer benchmark with a ViT-B/16 surrogate. The same estimator story survives the source-family swap: MFC-Std gains +5.4 over DeCoWA, and MFC-Full gains +2.8 over matched OPS. All entries are ASR (%) ↑.

Category	Attack	ViT-B/16 ↑	CNN targets			Transformer targets				Averages		
			RN50	VGG16	MN-v2	Inc-v3	PiT-B	Vis-S	Swin-T	CNN Avg. ↑	Transformer Avg. ↑	Black-box Avg. ↑
Gradient-based	VMI-FGSM	98.4	49.2	63.1	63.1	51.2	55.1	56.9	67.3	56.7	59.8	58.0
	GRA	97.0	65.6	72.4	72.8	66.9	71.2	71.7	78.2	69.4	73.7	71.3
Input-transformation-based	DIM	88.2	48.1	60.8	59.5	51.6	54.1	55.8	60.9	55.0	56.9	55.8
	SIM	99.2	51.4	66.0	64.7	53.5	57.0	58.8	70.7	58.9	62.2	60.3
	Admix	99.1	52.4	67.3	67.1	55.5	57.7	61.2	72.6	60.6	63.8	62.0
	SIA	97.3	75.2	78.9	81.5	71.8	81.4	81.2	84.2	76.8	82.3	79.2
	BSR	89.0	75.9	80.3	81.2	76.3	80.6	79.5	81.0	78.4	80.4	79.3
	DeCoWA	91.6	73.8	81.2	80.8	76.9	82.6	81.5	82.6	78.2	82.2	79.9
	OPS	98.7	91.1	93.6	94.1	93.2	95.3	94.6	95.2	93.0	95.0	93.9
	L2T	96.5	77.9	84.4	84.3	80.4	82.5	82.1	85.7	81.8	83.4	82.5
Ours	**MFC-Std**	**100.0**	80.1	86.5	85.5	83.4	87.3	86.5	87.8	83.9	87.2	85.3
Ours	**MFC-Full**	**100.0**	**94.1**	**96.5**	**95.5**	**95.4**	**98.3**	**98.5**	**98.8**	**95.4**	**98.5**	**96.7**

4.5 Diagnostic Mechanism Analysis Around the DeCoWA Backbone

The mechanism section focuses on one structured ablation table that studies composition, fusion rule, and per-family budget inside a reduced family-only estimator around the DeCoWA backbone.

4.5.1 Diagnostic Family Ablation. The next question is which parts of the reduced MFC-Std estimator matter most: the three-family design itself, the fusion rule, or the per-family warp budget. Table 3 reports all three blocks on the full 1000-image ResNet-50-source benchmark under that reduced family-only MFC-Std protocol.

Table 3 isolates how composition, fusion rule, and per-family budget shape the estimator. Within this reduced protocol, moving from one random family to two-family mixtures already lifts the black-box average from 75.7% to 80.7% and 79.6%, while the full three-family setting reaches 82.2%. Equal weights also outperform the front-loaded 0.8/0.1/0.1 mix (82.2% vs. 78.5%), and increasing the per-family budget from 4 to 10 and 20 warps raises the black-box average from 64.3% to 73.6% and 78.6%.

4.6 Fixed-Budget, Sampling, and Runtime Controls

The main tables establish a strong operating point, and the control block sharpens the attribution. Under an exact 60-view budget, the three-family 20×3 setting reaches 91.2% black-box ASR, outperforming a 60-random-warp control at 89.0% by 2.2 points. Keeping the same 20×3 decomposition, family-level consensus is also better than view-level averaging and view-level consensus by 1.6 and 1.1 points, respectively. The two-family controls (`random+equiangular` and `random+antithetic`) fall between the 60-random baseline and the full triplet, so the benefit is not carried by one extra family alone.

A second question is whether the effect can be explained as a generic low-discrepancy sampling trick in the released 2-D warp space. The strongest of six generic 2-D samplers—random-disk, Sobol, Halton, Latin, ring, and Poisson—reaches 90.5%, still 0.7 points below the three-family 20×3 setting. This comparison isolates the gain from a simple sequence-replacement explanation and reinforces the family-estimation view.

The matched runtime profile completes the same picture from the cost side. MFC-Std is 2.9× slower than DeCoWA in sec/img, with 120 versus 40 equivalent gradient evaluations per step and 30.0 per image, which is close to the intended 3× view budget. MFC-Full

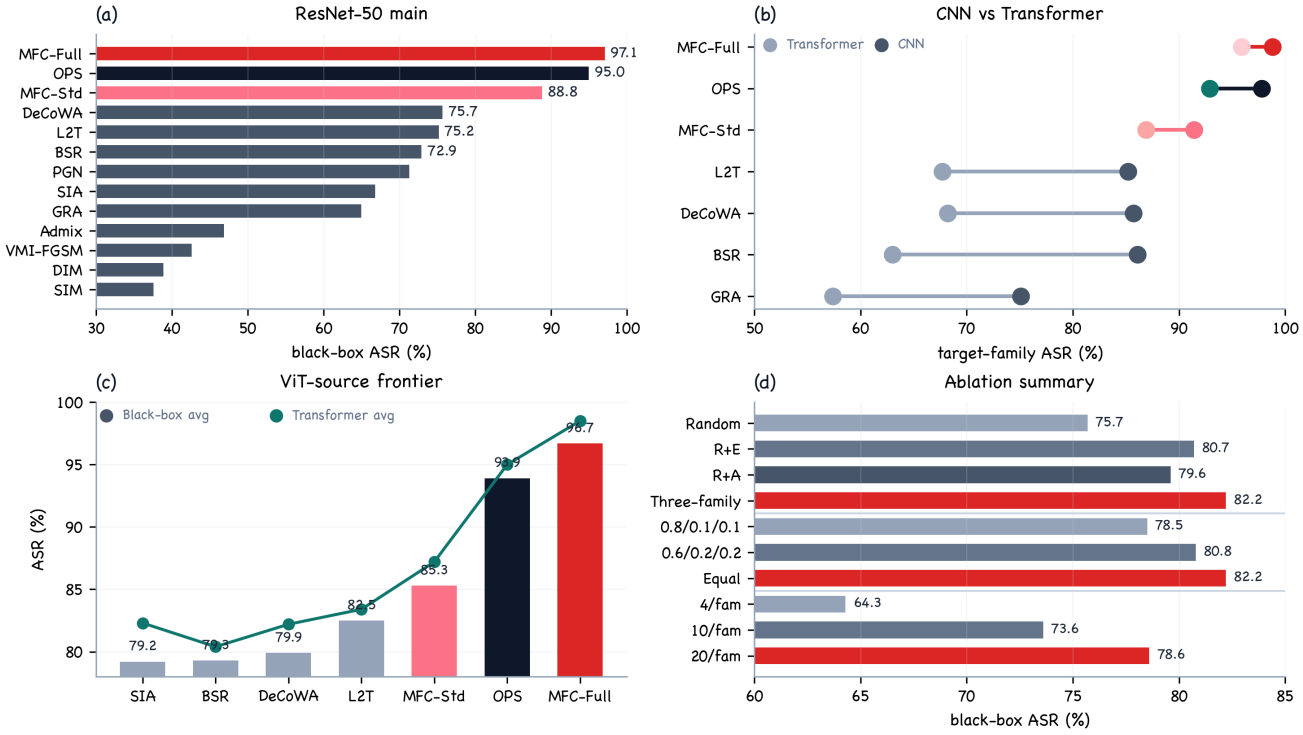


Figure 4: Main results and diagnostic summary. (a) Black-box-average ASR for the matched ResNet-50-source benchmark. (b) CNN-target and transformer-target averages for representative strong baselines together with MFC-Std and MFC-Full. (c) Black-box-average bars and transformer-target markers for the matched ViT-B/16-source benchmark. (d) Black-box-average summary of the family-composition, family-weight, and per-family warp-budget diagnostics.

Table 3: Diagnostic ablation of a reduced family-only MFC-Std variant on the 1000-image ResNet-50-source benchmark. Within this stripped-down setting, the full triplet is strongest among compositions and equal weights are strongest among fixed mixtures. All entries are ASR (%) ↑; the composition rows are within-design diagnostics rather than fixed-total-cost comparisons.

Study	Setting	RN50 ↑	CNN targets			Transformer targets				Averages		
			VGG16	MN-v2	Inc-v3	ViT-B/16	PiT-B	Vis-S	Swin-T	CNN Avg. ↑	Transformer Avg. ↑	Black-box Avg. ↑
Composition	Random only	94.5	87.3	88.0	82.8	53.5	66.7	76.1	75.2	86.0	67.9	75.7
Composition	Random + equiang.	96.9	92.1	91.1	87.0	57.8	74.1	81.9	81.1	90.1	73.7	80.7
Composition	Random + antithetic	97.2	90.4	90.9	85.3	57.9	72.9	81.5	78.6	88.9	72.7	79.6
Composition	**Three-family (R+E+A)**	**97.5**	**92.2**	**92.6**	**87.4**	**60.2**	**76.3**	**84.4**	**82.3**	**90.7**	**75.8**	**82.2**
Weights	**Equal (1/3, 1/3, 1/3)**	**97.9**	**92.3**	**92.1**	**87.6**	**59.9**	**76.4**	**84.4**	**82.8**	**90.7**	**75.9**	**82.2**
Weights	Main (0.8, 0.1, 0.1)	96.0	89.5	90.0	84.3	55.9	71.5	80.2	77.8	87.9	71.3	78.5
Weights	Balanced (0.6, 0.2, 0.2)	97.3	91.5	91.9	86.2	58.9	74.0	81.8	81.2	89.9	74.0	80.8
Warp budget	4 / family	85.8	79.4	76.3	72.0	40.8	55.3	63.0	63.5	75.9	55.7	64.3
Warp budget	10 / family	94.2	86.3	85.0	80.8	50.1	65.5	74.1	73.5	84.0	65.8	73.6
Warp budget	**20 / family**	**96.9**	**89.5**	**90.2**	**84.9**	**55.8**	**70.9**	**80.6**	**78.5**	**88.2**	**71.5**	**78.6**

is 2.8× slower than OPS, with 2703 versus 901 equivalent gradient evaluations per step, and stays within a 4.2 GB peak-memory profile.

Taken together, these controls answer three different attribution questions. The fixed-budget comparison asks whether family structure still helps when total views are matched exactly; the sampler sweep asks whether the gain is merely a better way to tile the released 2-D disk; and the runtime profile asks whether the realized cost stays close to the intended view multiplier. All three checks support the same conclusion: the completed triplet row is better explained by family-level estimation than by a bookkeeping artifact.

5 Discussion

The clearest reading of the paper is also the most useful one. In the released 3×3 TPS setting, family-level estimation is materially stronger than the single-family DeCoWA estimator, and the same factorization remains beneficial after the outer search is strengthened by OPS. The contribution is therefore an estimator redesign with a primary DeCoWA-backed claim, a secondary OPS-backed

Table 4: Complementary controls on the completed 1000-image ResNet-50-source benchmark. The three-family estimator stays ahead under matched budget, generic 2-D sampling, and runtime checks.

Control	Result
60-random-warp baseline	89.0% black-box ASR
Three-family 20×3	91.2% (+2.2 vs. random60)
Family consensus vs. view average	+1.6 points
Family consensus vs. view consensus	+1.1 points
Best generic 2-D sampler	90.5% (−0.7 vs. triplet)
MFC-Std runtime vs. DeCoWA	2.9× sec/img; 3.0 eq./step
MFC-Full runtime vs. OPS	2.8× sec/img; 3.0 eq./step
MFC-Full peak memory	$\leq$ 4200 MB

generalization check, and a matched fixed-budget advantage inside the same low-dimensional warp setting. More broadly, the paper suggests that structured transfer attacks should be designed around meaningful estimation units rather than treating every sampled view as interchangeable. In our case, the natural unit is the deformation family: stochastic variability is reduced within a family first, and only then is directional diversity fused across families.

Table 1 and Figure 3 show that MFC-Std gains concentrate under stronger source–target mismatch, especially when a CNN surrogate attacks transformer-like victims, which makes the estimator story more specific than a generic strengthening trick.

This mismatch pattern is important for how the method should be interpreted. A generic strengthening trick would usually lift all targets more uniformly, but MFC-Std instead delivers its clearest gains exactly where the surrogate-target gap is larger. That is consistent with the estimator view developed in Sec. 3: when the transferable direction is harder to expose, organizing deformation space into complementary families matters more than repeatedly sampling one stochastic family. The ViT-B/16-source block then provides the same story under a source swap, showing that the effect is not tied to one surrogate architecture.

At the same time, the released 3×3 setting is the public DeCoWA operating point that practitioners actually inherit from the released codebase, so showing a cleaner estimator advantage there is already useful in practice.

The results also suggest a practical reporting lesson for future transfer-attack benchmarks. When a method expands the view pool, reporting only the total number of sampled views is not enough: the partition of those views into families, and the level at which gradients are fused, can change transfer behavior materially even under the same nominal budget. Our fixed-budget controls make this concrete by showing that `random60` and `triplet20x3` are not interchangeable once estimator structure is taken seriously. This point matters beyond the current method name. Once view sets are structured, the benchmark should make that structure visible, because two attacks with the same nominal number of views can allocate those views very differently and therefore behave very differently in transfer.

6 Conclusion

This paper studies transferable black-box attacks through family-level estimation in low-dimensional deformation space. Under the released DeCoWA 3×3 TPS mesh, the key observation is that stochastic warp views should first be stabilized within families and only then mixed across complementary families. This yields a strong controlled operating point on the DeCoWA backbone and a consistent OPS-backed generalization, with the clearest gains under cross-architecture mismatch and most notably when a CNN surrogate attacks transformer-like victims. Complementary fixed-budget and sampling controls show that the gain persists under matched computation, including a +2.2 margin at matched 60 views, so it is not reducible to more surrogate-side search.

More broadly, the results suggest that transfer gain does not increase monotonically with attack-pipeline complexity. Within a well-studied backbone, a principled but minimal estimator change can expose a clearer transfer improvement than many outer-pipeline extensions, and the matched-budget controls rule out attributing that gain to extra computation. Although random, equiangular, and antithetic sampling are each classical techniques, combining them as family-level estimation for deformation-constrained transfer is, to our knowledge, first explored here; prior deformation-based attacks including DeCoWA, OPS, BSR, SIA, and L2T did not examine this design axis. If family-level estimation proves similarly effective in other view-based attacks such as BSR or SIA, it may point to a more general principle for structured view aggregation.

References

[1] Bin Chen, Jiali Yin, Shukai Chen, Bohao Chen, and Ximeng Liu. 2023. An adaptive model ensemble adversarial attack for boosting adversarial transferability. In *Proceedings of the IEEE/CVF international conference on computer vision*. 4489–4498.

[2] Yinpeng Dong, Tianyu Pang, Hang Su, and Jun Zhu. 2019. Evading defenses to transferable adversarial examples by translation-invariant attacks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4312–4321.

[3] Zhengwei Fang, Rui Wang, Tao Huang, and Liping Jing. 2024. Strong transferable adversarial attacks via ensembled asymptotically normal distribution learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 24841–24850.

[4] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. 2014. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572* (2014).

[5] Yu Guo, Weiquan Liu, Qingshan Xu, Shijun Zheng, Shujun Huang, Yu Zang, Siqi Shen, Chenglu Wen, and Cheng Wang. 2025. Boosting adversarial transferability through augmentation in hypothesis space. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 19175–19185.

[6] Qian Huang, Isay Katsman, Horace He, Zeqi Gu, Serge Belongie, and Ser-Nam Lim. 2019. Enhancing adversarial example transferability with an intermediate level attack. In *Proceedings of the IEEE/CVF international conference on computer vision*. 4733–4742.

[7] Shixin Li, Chaoxiang He, Xiaojing Ma, Bin Benjamin Zhu, Shuo Wang, Hongsheng Hu, Dongmei Zhang, and Linchen Yu. 2025. Enhancing Adversarial Transferability with Checkpoints of a Single Model's Training. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 20685–20694.

[8] Kaisheng Liang and Bin Xiao. 2023. Styless: Boosting the transferability of adversarial examples. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8163–8172.

[9] Jiadong Lin, Chuanbiao Song, Kun He, Liwei Wang, and John E Hopcroft. 2019. Nesterov accelerated gradient and scale invariance for adversarial attacks. *arXiv preprint arXiv:1908.06281* (2019).

[10] Qinliang Lin, Cheng Luo, Zenghao Niu, Xilin He, Weicheng Xie, Yuanbo Hou, Linlin Shen, and Siyang Song. 2024. Boosting adversarial transferability across model genus by deformation-constrained warping. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 3459–3467.

[11] Yanpei Liu, Xinyun Chen, Chang Liu, and Dawn Song. 2016. Delving into transferable adversarial examples and black-box attacks. *arXiv preprint arXiv:1611.02770* (2016).

[12] Yantao Lu, Yunhan Jia, Jianyu Wang, Bai Li, Weiheng Chai, Lawrence Carin, and Senem Velipasalar. 2020. Enhancing cross-task black-box transferability of adversarial examples with dispersion reduction. In *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*. 940–949.

[13] Thibault Maho, Seyed-Mohsen Moosavi-Dezfooli, and Teddy Furon. 2023. How to choose your best allies for a transferable attack?. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4542–4551.

[14] Jinjia Peng, Zeze Tao, Huibing Wang, Meng Wang, and Yang Wang. 2025. Boosting Adversarial Transferability via Residual Perturbation Attack. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 1261–1270.

[15] Yunxiao Qin, Yuanhao Xiong, Jinfeng Yi, and Cho-Jui Hsieh. 2023. Training meta-surrogate model for transferable adversarial attack. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 37. 9516–9524.

[16] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. 2015. Imagenet large scale visual recognition challenge. *International journal of computer vision* 115, 3 (2015), 211–252.

[17] Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. 2013. Intriguing properties of neural networks. *arXiv preprint arXiv:1312.6199* (2013).

[18] Bowen Tang, Zheng Wang, Yi Bin, Qi Dou, Yang Yang, and Heng Tao Shen. 2024. Ensemble diversity facilitates adversarial transferability. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 24377–24386.

[19] Trustworthy-AI-Group. 2024. TransferAttack: A PyTorch Framework to Boost the Adversarial Transferability for Image Classification. https://github.com/Trustworthy-AI-Group/TransferAttack.

[20] Kunyu Wang, Xuanran He, Wenxuan Wang, and Xiaosen Wang. 2024. Boosting adversarial transferability by block shuffle and rotation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 24336–24346.

[21] Xiaosen Wang and Kun He. 2021. Enhancing the transferability of adversarial attacks through variance tuning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 1924–1933.

[22] Xiaosen Wang, Xuanran He, Jingdong Wang, and Kun He. 2021. Admix: Enhancing the transferability of adversarial attacks. In *Proceedings of the IEEE/CVF international conference on computer vision*. 16158–16167.

[23] Xiaosen Wang, Zeliang Zhang, and Jianping Zhang. 2023. Structure invariant transformation for better adversarial transferability. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4607–4619.

[24] Zhibo Wang, Hengchang Guo, Zhifei Zhang, Wenxin Liu, Zhan Qin, and Kui Ren. 2021. Feature importance-aware transferable adversarial attacks. In *Proceedings of the IEEE/CVF international conference on computer vision*. 7639–7648.

[25] Zhipeng Wei, Jingjing Chen, Zuxuan Wu, and Yu-Gang Jiang. 2023. Enhancing the self-universality for transferable targeted attacks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 12281–12290.

[26] Dongxian Wu, Yisen Wang, Shu-Tao Xia, James Bailey, and Xingjun Ma. 2020. Skip connections matter: On the transferability of adversarial examples generated with resnets. *arXiv preprint arXiv:2002.05990* (2020).

[27] Tao Wu, Tie Luo, and Donald C Wunsch II. 2024. LRS: Enhancing adversarial transferability through Lipschitz regularized surrogate. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 6135–6143.

[28] Weibin Wu, Yuxin Su, Xixian Chen, Shenglin Zhao, Irwin King, Michael R Lyu, and Yu-Wing Tai. 2020. Boosting the transferability of adversarial samples via attention. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 1161–1170.

[29] Cihang Xie, Zhishuai Zhang, Yuyin Zhou, Song Bai, Jianyu Wang, Zhou Ren, and Alan L Yuille. 2019. Improving transferability of adversarial examples with input diversity. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2730–2739.

[30] Yifeng Xiong, Jiadong Lin, Min Zhang, John E Hopcroft, and Kun He. 2022. Stochastic variance reduced ensemble adversarial attack for boosting the adversarial transferability. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 14983–14992.

[31] Wen Yang, Guodong Liu, and Di Ming. 2025. SMP-Attack: Boosting the Transferability of Feature Importance-based Adversarial Attack with Semantics-aware Multi-granularity Patchout. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4444–4454.

[32] Jianping Zhang, Jen-tse Huang, Wenxuan Wang, Yichen Li, Weibin Wu, Xiaosen Wang, Yuxin Su, and Michael R Lyu. 2023. Improving the transferability of adversarial samples by path-augmented method. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8173–8182.

[33] Zhengyu Zhao, Hanwei Zhang, Renjue Li, Ronan Sicre, Laurent Amsaleg, Michael Backes, Qi Li, and Chao Shen. 2023. Revisiting transferable adversarial image examples: Attack categorization, evaluation guidelines, and new insights. *arXiv preprint arXiv:2310.11850* (2023).

[34] Hegui Zhu, Yuchen Ren, Xiaoyan Sui, Lianping Yang, and Wuming Jiang. 2023. Boosting adversarial transferability via gradient relevance attack. In *Proceedings of the IEEE/CVF international conference on computer vision*. 4741–4750.

[35] Rongyi Zhu, Zeliang Zhang, Susan Liang, Zhuo Liu, and Chenliang Xu. 2024. Learning to transform dynamically for better adversarial transferability. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 24273–24283.